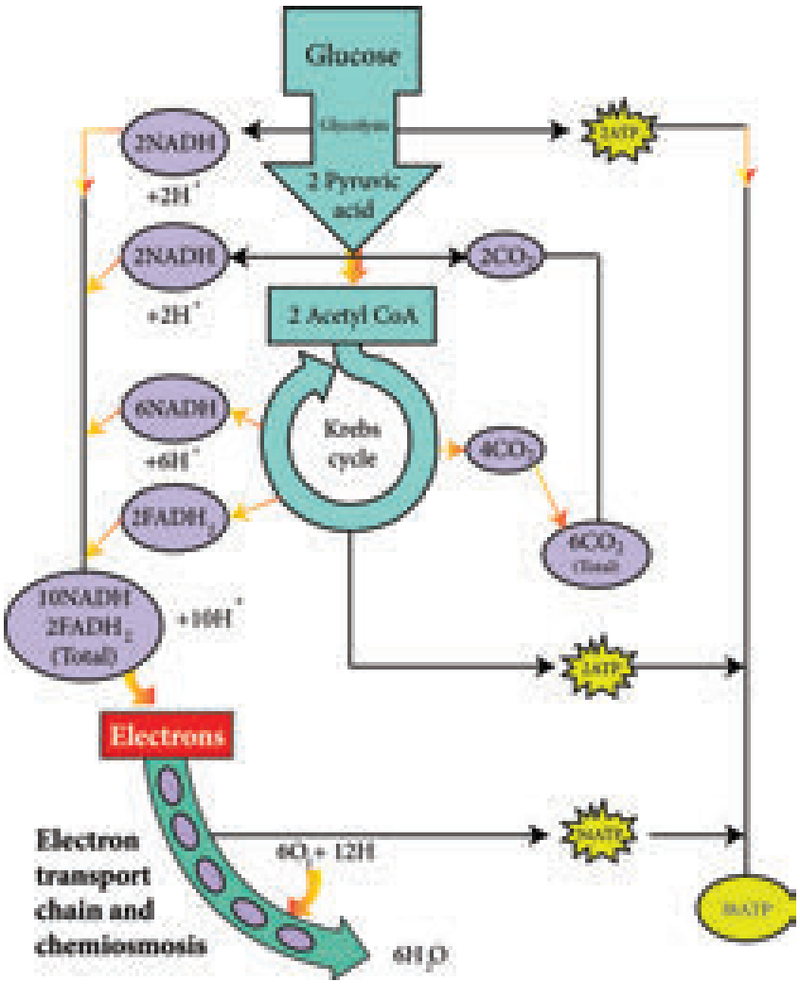


Figure 4.7: Overview of aerobic respiration

Table 4.2: Net gain of ATP produced during aerobic respiration of glucose in prokaryotes

Glycolysis 1. Oxidation of glucose to Pyruvic acid. 2. Production of 2 NADH	**Preparatory step** 2 ATP (substrate level phosphorylation) 6 ATP (Oxidative phosphorylation in ETC)
Preparatory step 1. Formation of acetyl CoA produces 2NADH	6 ATP (Oxidative phosphorylation in ETC)
Krebs cycle 1. Oxidation of succinyl CoA to succinic acid 2. Production of 6 NADH 3. Production of 2 FADH 4. Oxidative phosphorylation with 2 NADH	2 ATP (Substrate level phosphorylation) 18 ATP (Oxidative phosphorylation in ETC) 4 ATP (Oxidative phosphorylation in ETC) 6 ATP **Total 38 ATP**

1 NADH = 3 ATPs and 1$FADH_2$ = 2 ATP

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